

Meta-Learning-Driven Joint Power Allocation and Phase-Shift Design for RIS-Assisted NOMA-ISAC Systems

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Abstract—This letter investigates a reconfigurable intelligent surface (RIS)-assisted multiple-input single-output (MISO) integrated sensing and communication (ISAC) system, where a base station (BS) simultaneously serves a near and a far user via non-orthogonal multiple access (NOMA) while illuminating a sensing target. We aim to maximize a weighted communication-and-sensing throughput by jointly optimizing the NOMA power allocation and the RIS phase shifts, subject to per-link communication and sensing quality-of-service (QoS) constraints. To tackle the formulated non-convex joint optimization problem, a meta-learning (MAML)-based deep-learning (DL) algorithm is proposed, in which a lightweight network predicts the power allocation from channel features while the RIS phase shifts are refined per channel realization by an unrolled inner optimizer, and second-order meta-gradients couple the two stages to improve generalization. Simulation results demonstrate that the proposed algorithm drives the joint QoS-violation rate from 29.4% to 0.1%, scales the weighted sum-rate by about 65% as the RIS grows from 8 to 64 elements, and consistently outperforms the orthogonal multiple access (OMA), no-RIS, and partially-optimized baselines.

Index Terms—Reconfigurable intelligent surface (RIS), integrated sensing and communication (ISAC), non-orthogonal multiple access (NOMA), deep learning, meta-learning (MAML), MISO.

I. INTRODUCTION

INTEGRATED sensing and communication (ISAC) has emerged as a key enabler for next-generation wireless networks, jointly designing the sensing and communication functions on a shared spectral and spatial resource to improve spectrum, energy, and hardware efficiency [1], [2]. When combined with non-orthogonal multiple access (NOMA), which superposes users in the power domain and separates them by successive interference cancellation (SIC), ISAC can further raise the throughput of dense Internet-of-Things deployments. However, severe blockage and path loss often degrade both functions, motivating the use of a reconfigurable intelligent surface (RIS) to reshape the propagation environment through low-cost passive elements [3].

RIS-assisted ISAC resource management has attracted growing interest. In [4], a manifold-based scheme jointly optimized hybrid beamforming and RIS phase shifts to shape the ISAC beampattern. In [5], a penalty-based block-coordinate-descent method minimized the Cramér–Rao bound for RIS-aided

mmWave ISAC. A rate-splitting RIS-ISAC system was studied in [6], where beamforming, rate-splitting, and RIS phases were jointly designed to maximize throughput. For RIS-NOMA networks, [7] introduced a deep-learning (DL) framework with model-agnostic meta-learning (MAML) to jointly optimize the BS power allocation and RIS phase shifts. Most of these designs, however, rely on per-instance alternating optimization (AO), which incurs high computational complexity and is ill-suited to the short channel-coherence times of practical links; learning the high-dimensional RIS phase *directly* with a network, on the other hand, tends to overfit and to generalize poorly across channel realizations [7].

To overcome these challenges, we propose a MAML-based DL framework for a MISO RIS-NOMA-ISAC downlink that *factorizes* the joint design according to its dimensionality. The main contributions are summarized as follows. (i) We formulate a weighted communication-plus-sensing throughput maximization problem with explicit per-link QoS constraints, and contrast it with an OMA baseline. (ii) We propose a bilevel solver in which a small network predicts the low-dimensional NOMA power split—the quantity that benefits from learned priors—while the high-dimensional RIS phase is refined per channel realization by an unrolled inner loop, with second-order meta-gradients linking the two; this factorization is the key to stable training and generalization. (iii) Simulation results quantify the resulting NOMA, RIS, and learning gains and confirm the ordering NOMA-RIS > NOMA > OMA-RIS > OMA.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a downlink MISO RIS-ISAC system in which a BS equipped with M antennas, aided by an RIS of N elements, simultaneously serves a near user (n) and a far user (f), each with a single antenna, and illuminates a point sensing target (T). The RIS applies a diagonal phase-shift matrix $\Phi = \text{diag}(e^{j\phi_1}, \dots, e^{j\phi_N})$ with unit-modulus coefficients $\phi_i \in [0, 2\pi)$.

1) *Channel Model*: With $\mathbf{G} \in \mathbb{C}^{N \times M}$ the BS–RIS channel and $\mathbf{r}_k \in \mathbb{C}^N$ the RIS–user channel, the effective downlink channel of user $k \in \{n, f\}$ is the cascade

$$\mathbf{h}_k = \mathbf{G}^H \Phi \mathbf{r}_k \in \mathbb{C}^M. \quad (1)$$

In the RIS-aided case the direct BS–user path is obstructed and the cascade (1) dominates; the *no-RIS* baseline retains only the

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(weak) direct link. The round-trip sensing response toward the target is captured by the effective matrix $\mathbf{G}_T \in \mathbb{C}^{M \times M}$ with gain β_T ; since $\mathbf{G}_T$ is rank-one, the optimal sensing receive combiner $\mathbf{u}_T$ is obtained in closed form.

2) *NOMA Communication and Sensing Model*: The BS transmits the superposed signal

$$\mathbf{x} = (\sqrt{a_n P} s_n + \sqrt{a_f P} s_f) \mathbf{w}_c + \sqrt{a_T P} \mathbf{w}_T s_T, \quad (2)$$

where P is the transmit power, $a_n + a_f + a_T = 1$ is the power split over the near/far/sensing streams, $\mathbf{w}_c = \mathbf{h}_f / \|\mathbf{h}_f\|$ is the *cluster* beamformer (matched to the weaker user and shared by the superposed streams), and $\mathbf{w}_T$ is the sensing beamformer placed in the null space of $[\mathbf{h}_n, \mathbf{h}_f]$ to limit leakage onto the users. With $g_{kj} = |\mathbf{h}_k^H \mathbf{w}_j|^2$ and perfect SIC at the near user, the SINRs of the far user, near user, and the sensing link are

$$\gamma_f = \frac{a_f g_{ff} P}{a_n g_{ff} P + a_T g_{fT} P + \sigma^2}, \quad (3)$$

$$\gamma_n = \frac{a_n g_{nn} P}{a_T g_{nT} P + \sigma^2}, \quad (4)$$

$$\gamma_s = \frac{a_T |\mathbf{u}_T^H \mathbf{G}_T \mathbf{w}_T|^2 P}{\sigma^2}, \quad (5)$$

where σ^2 is the noise power. The achievable rates are $R_k = \log_2(1 + \gamma_k)$, $k \in \{n, f, s\}$, and the weighted communication-and-sensing throughput is

$$R_{DL} = \omega_c (R_n + R_f) + \omega_s R_s, \quad (6)$$

with weights $\omega_c = 0.7$ and $\omega_s = 0.3$. As a benchmark, the *OMA* scheme serves the same streams over orthogonal time fractions $t_n + t_f + t_T = 1$ with per-user matched beamforming, giving $R_k^{\text{OMA}} = t_k \log_2(1 + \|\mathbf{h}_k\|^2 P / \sigma^2)$ for the users; it trades the inter-user interference and SIC ceiling of NOMA for a pre-log time penalty.

B. Problem Formulation

We jointly optimize the power split $\mathbf{a} = (a_n, a_f, a_T)$ and the RIS phases $\boldsymbol{\phi} = (\phi_1, \dots, \phi_N)$ to maximize the weighted throughput subject to per-link QoS, yielding problem (P1):

$$(P1) \quad \max_{\mathbf{a}, \boldsymbol{\phi}} \quad R_{DL} \quad (7a)$$

$$\text{s.t.} \quad R_n + R_f \geq R_{\text{th},c}, \quad (7b)$$

$$R_s \geq R_{\text{th},s}, \quad (7c)$$

$$a_n + a_f + a_T = 1, \quad a_k \geq 0, \quad (7d)$$

$$|e^{j\phi_i}| = 1, \quad i = 1, \dots, N. \quad (7e)$$

Constraint (7b) enforces the communication sum-rate floor and (7c) the sensing-rate floor; (7d) is the power simplex and (7e) the unit-modulus phase constraint. Problem (P1) is non-convex in both the simplex-constrained power split and the unit-modulus phases, which are moreover coupled through the cascade (1).

III. PROPOSED MAML-BASED SOLUTION

The central design choice is *what to learn*. The RIS phase vector $\boldsymbol{\phi} \in \mathbb{R}^N$ is high-dimensional and channel-specific, so learning it directly invites overfitting; the power split $\mathbf{a}$ is low-dimensional and admits a stable channel-to-allocation map. We therefore split the variables across two nested loops and meta-train only the power map.

A. Feature Map and Power-Allocation Network

From the effective gains we form a compact feature vector $\mathbf{z}$ (per-link gains in dB, the QoS margins, and an SNR-budget term), which a small multilayer perceptron f_θ maps to the power split through a softmax head, $\mathbf{a} = \text{softmax}(f_\theta(\mathbf{z}))$, thereby satisfying the simplex constraint (7d) by construction.

B. MAML-Based Bilevel Optimization

1) *Inner Loop – Per-Channel RIS Phase Refinement*: For each channel realization the phases are refined by K steps of projected gradient ascent on the objective,

$$\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} + \eta \nabla_{\boldsymbol{\phi}} R_{DL}(\mathbf{a}, \boldsymbol{\phi}), \quad |e^{j\phi_i}| = 1, \quad (8)$$

with step size η . This per-instance refinement is what a one-shot network cannot reliably reproduce, and it keeps the learned object low-dimensional.

2) *Outer Loop – Meta-Training*: The network weights $\boldsymbol{\theta}$ are meta-trained across a batch of channels with the differentiable Lagrangian

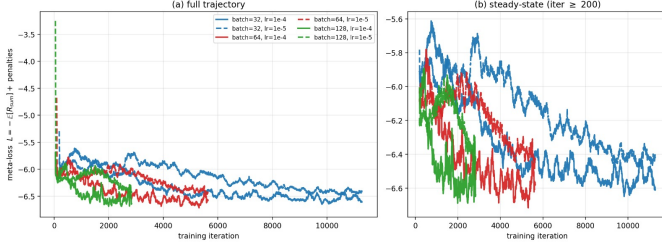
$$\mathcal{L}(\boldsymbol{\theta}) = -\mathbb{E}[R_{DL}] + \lambda_c \mathbb{E}[(R_{\text{th},c} - R_n - R_f)_+] + \lambda_s \mathbb{E}[(R_{\text{th},s} - R_s)_+], \quad (9)$$

where $(\cdot)_+ = \max(\cdot, 0)$ penalizes QoS violations and λ_c, λ_s are penalty factors. Crucially, the meta-gradient is taken *through* the unrolled inner loop (8) (second-order MAML), so $\boldsymbol{\theta}$ learns power priors that are good *after* phase refinement.

3) *Training Procedure*: At each iteration a mini-batch of channels is drawn; for each channel the inner loop (8) refines $\boldsymbol{\phi}$ from a random initialization while the network predicts $\mathbf{a}$, the loss (9) is accumulated, and $\boldsymbol{\theta}$ is updated by Adam using the second-order meta-gradient. A held-out validation set selects the best checkpoint. Because the learned object is the low-dimensional power map, the validation throughput tracks—indeed slightly exceeds—the training throughput, i.e. the model generalizes rather than memorizes.

C. Complexity

Training is performed offline; the cost per meta-iteration scales as $\mathcal{O}(BK(MN + P_\theta))$ with batch size B , inner steps K , and P_θ network parameters. The online stage requires only one forward pass of f_θ plus the K lightweight inner steps, so the main burden is shifted to offline training—unlike AO, which re-solves the joint problem for every channel realization.

Fig 4 — NOMA MAML training loss vs iterations (batch x lr sweep, $M=4$ fair, $N=8$)

 Fig. 1. MAML meta-loss versus training iteration over a batch/learning-rate sweep ($M=4$, $N=8$). (a) Full trajectory; (b) steady state (iter ≥ 200).

IV. NUMERICAL RESULTS AND ANALYSIS

Unless otherwise stated, we set $M = 4$ antennas, $N = 8$ RIS elements, weights $\omega_c/\omega_s = 0.7/0.3$, and physical noise σ^2 ; the QoS thresholds $R_{th,c}$ and $R_{th,s}$ are calibrated to the operating regime, and the transmit power is swept over 5–25 dBm. The inner loop uses $K=5$ steps; the network is trained with Adam over a batch/learning-rate sweep. We compare the following schemes:

- **NOMA-RIS (proposed)** and **NOMA (no-RIS)**: superposition with SIC, with and without the RIS.
- **OMA-RIS** and **OMA (no-RIS)**: orthogonal time-division baseline with per-user beamforming.
- **Ablation**: randPhi_randAlpha (no optimization), randPhi_fairAlpha (fixed equal split), randPhi_optAlpha (learned power only), randAlpha_optPhi (optimized phase only), and fullOptimized (proposed).

Fig. 1 shows the MAML meta-loss (9) over training iterations for several batch/learning-rate settings. The loss decreases smoothly and the weighted sum-rate climbs steadily, with no epoch-one collapse, indicating stable second-order meta-optimization; the steady-state panel confirms continued, generalizing improvement.

Fig. 2 plots the empirical CDF of the weighted sum-rate R_{DL} over 10^4 test channels for the ablation. Joint optimization of phase and power (fullOptimized) stochastically dominates the partial designs, raising the median from ~ 5.0 to ~ 6.7 bps/Hz; phase optimization accounts for most of the rate gain, while the learned power split adds a further, QoS-critical improvement. This is reflected in Fig. 3: the joint QoS-violation rate falls monotonically from 29.4% (no optimization) to 0.1% (proposed), with both the communication (q_c) and sensing (q_s) constraints satisfied almost surely.

Fig. 4 reports R_{sum} versus the RIS size $N \in \{8, 16, 32, 64\}$ at $P = 15$ dBm. The rate grows monotonically with N for both access schemes—from 6.9 to 11.4 bps/Hz for NOMA and from 4.1 to 7.3 bps/Hz for OMA—and NOMA retains an $\sim 65\%$ advantage throughout, showing that the passive aperture and the access scheme contribute complementary gains.

Finally, Fig. 5 sweeps the transmit power. Panel (a) establishes the ordering NOMA-RIS $>$ NOMA $>$ OMA-RIS $>$ OMA across the whole range: NOMA’s full-time superposition outperforms OMA’s orthogonal slots, and the RIS adds a consistent margin on top of each access scheme (e.g. a $\sim 27\%$ RIS gain for NOMA at $P = 15$ dBm). Panel (b) shows the

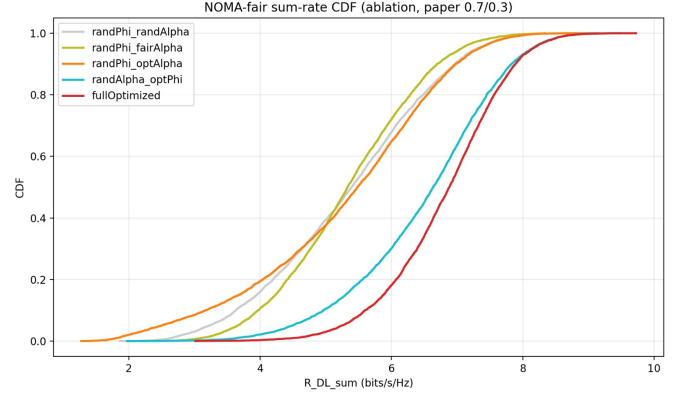
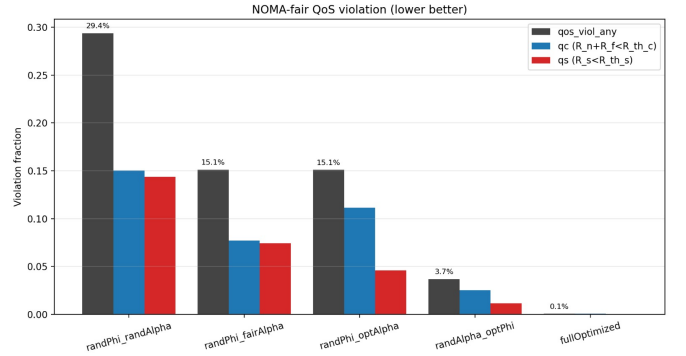

 Fig. 2. CDF of the weighted sum-rate R_{DL} over the test set for the five ablation configurations.


Fig. 3. QoS-violation fraction (lower is better). The joint violation rate drops from 29.4% to 0.1% along the ablation.

corresponding QoS-violation rate: NOMA-RIS reaches near-zero violation already at low power, whereas the no-RIS and OMA schemes need substantially more power to become feasible, quantifying the role of the RIS in jointly meeting the sensing and communication QoS.

V. CONCLUSION

In this letter, a MISO RIS-assisted NOMA-ISAC downlink was investigated. To maximize a weighted communication-and-sensing throughput under per-link QoS constraints, we developed a MAML-based DL framework that factorizes the joint design into a learned low-dimensional power allocation and an unrolled per-channel RIS phase optimization, coupled by second-order meta-gradients. Simulation results show that the proposed scheme trains stably and generalizes, meets the QoS almost surely (0.1% violation), scales with the RIS aperture, and consistently outperforms the OMA and no-RIS baselines. Future work includes multi-cluster NOMA, imperfect SIC/CSI, and a sensing metric beyond the rate surrogate (e.g. the Cramér–Rao bound).

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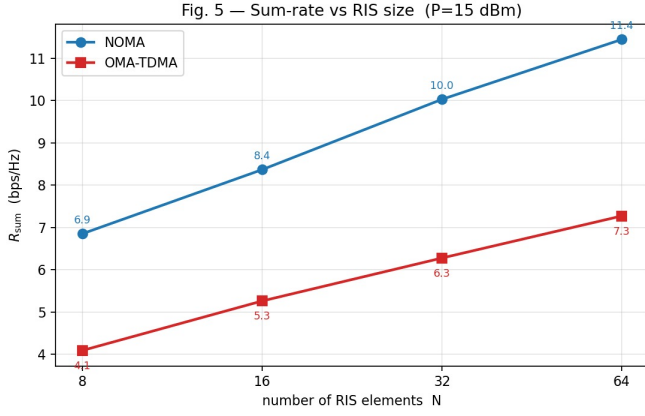


Fig. 4. Weighted sum-rate versus RIS size N at $P = 15$ dBm. NOMA dominates OMA at every aperture; both scale with N .

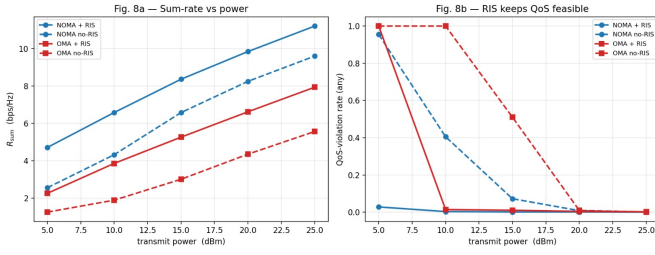


Fig. 5. (a) Weighted sum-rate and (b) joint QoS-violation rate versus transmit power for the four schemes.

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